

Detecting Customer Churn Risk from App Store Reviews: A Review-Mining Approach to Early Retention Signals

1. Problem Statement and Objectives

Problem Statement

Online retail platforms depend on repeat customers for sustainable revenue, yet many users disengage silently after a poor experience, well before this shows up in transactional revenue data. By the time falling purchase frequency reveals a churn problem, the customer relationship is often already unrecoverable. However, dissatisfied users frequently leave early warning signals in public app store reviews — complaints about delivery delays, refund disputes, counterfeit products, or poor customer service — long before they formally churn. This study treats app store reviews as a leading indicator of churn risk, mined directly from real, user-generated text rather than a pre-packaged transactional churn dataset.

Importance

Acquiring a new customer typically costs several times more than retaining an existing one, so an early, low-cost signal of dissatisfaction has direct commercial value. Understanding which complaint themes and behavioural patterns precede churn also surfaces broader operational issues — in logistics, refund handling, or product authenticity — that may be affecting the wider customer base, not just the reviewer.

Objectives

- Collect and analyse a real-world dataset of e-commerce app reviews via web scraping, across multiple marketplace apps, to identify behavioural and textual signals of churn risk.
- Build and evaluate machine learning classifiers (Logistic Regression, Random Forest) to automatically flag reviews indicating churn risk, using features that are genuinely independent of how the risk label itself was constructed.
- Segment reviews into behavioural clusters using unsupervised learning to identify distinct reviewer personas.
- Benchmark the review-mining approach against published churn-prediction and app-review-mining studies.
- Translate findings into concrete, actionable retention recommendations for marketing, customer relationship, and product teams.

2. Data Collection and Dataset Description

Data Collection Method

Data was collected through web scraping of Google Play Store reviews for major Indian e-commerce marketplace apps. No ready-made dataset from Kaggle, UCI, or a similar repository was used; all reviews represent genuine, real user-submitted text. Reviews were pulled per app and enriched with derived analytical fields (sentiment score, churn-risk flag, complaint theme cluster) to support downstream modelling.

App	Category	Approx. Churn-Risk Rate
Amazon	Marketplace	31.1%
ShopClues	Marketplace	21.2%
Ajio	Fashion	14.5%

App	Category	Approx. Churn-Risk Rate
Nykaa	Beauty	13.6%
Snapdeal	Value Marketplace	11.1%
Myntra	Fashion	10.4%
Meesho	Social Commerce	8.5%
Flipkart	Marketplace	5.1%

Total reviews scraped: 12,800 across 8 apps.

Dataset Description

The final cleaned dataset contains **12,800 reviews across 21 attributes**, comfortably above the 10,000-record target set in the original proposal.

Attribute	Description
review_id	Anonymized unique identifier for the review
app_name, app_category	Which app and category the review belongs to
rating, rating_bucket	Star rating (1-5) and a binned version
review_text	Raw review text
review_length, word_count	Text length metrics
review_date, review_day_of_week	When the review was posted
thumbs_up_count	Helpfulness votes from other users
app_version	App version at time of review (82.8% populated)
reply_content, has_reply	Whether/how the developer responded (66.1% have a reply)
sentiment_score, sentiment_label	Derived lexicon-based sentiment
churn_risk	Binary label: 1 = review indicates the user is at risk of leaving
explicit_churn_mention	1 if the review explicitly mentions uninstalling / switching apps
churn_signal	Derived flag combining rating, sentiment and explicit mentions
theme_cluster, complaint_theme	Numeric cluster id and human-readable complaint category

Privacy: review_id values are anonymized UUIDs; no reviewer names, emails, or device identifiers are retained in the dataset.

3. Data Preparation and Exploratory Analysis

Data Cleaning

- De-duplicated on review_id (no duplicates found in the 12,800-row scrape).
- Parsed review_date into a proper datetime and derived month / day-of-week features.
- Missing app_version (17.2% of rows) and reply_content (33.9% of rows) were treated as legitimately absent — older reviews predate version tagging, and most reviews never receive a developer reply — and flagged with explicit indicator columns rather than dropped.
- Basic text cleaning of review_text (lowercased, URLs and punctuation stripped) ahead of TF-IDF text analysis.

Key Descriptive Statistics

Overall churn-risk rate across all reviews: **14.5%** (1,856 of 12,800 reviews flagged at-risk). Ratings are heavily skewed toward 5 stars (a typical Play Store pattern), while sentiment shows a distinctly bimodal shape, clustering near strongly positive and mildly negative values rather than spreading smoothly — consistent with a lexicon-based scoring approach.

Complaint Theme Distribution (At-Risk Reviews)

Complaint Theme	Review Count	Share of At-Risk Reviews
Returns / Refunds & Support	1,163	62.7%
Delivery Issues	319	17.2%
Fake Products / Fraud	130	7.0%
Non-English Review	125	6.7%
Product Quality / Value	72	3.9%
Poor Service Quality	46	2.5%

Interpretation: Returns/refunds and support handling dominate churn-risk reviews, accounting for nearly two-thirds of all at-risk cases — far ahead of delivery issues (17.2%) or product authenticity concerns (7.0%). This single theme is the highest-leverage target for retention intervention.

Text Analysis (TF-IDF)

TF-IDF analysis comparing at-risk and not-at-risk review language shows a clear, intuitive separation. At-risk reviews are dominated by terms such as *bad*, *worst*, *delivery*, *service*, *customer*, *order*, *return*, *refund*, *poor*, and *support*; not-at-risk reviews are dominated by *good*, *nice*, *best*, *excellent*, *super*, and *quality*. A direct keyword check for explicit churn language (e.g. "uninstall", "switch to", "never using") found only 60 reviews (0.5% of the dataset) containing such phrasing, with 99.4% agreement against the explicit_churn_mention label — confirming that most churn risk is expressed indirectly through complaint content and sentiment rather than explicit uninstall statements.

Correlation and Label Construction Check

A correlation analysis of numeric features against churn_risk revealed an important methodological finding: rating ($r = -0.66$) and sentiment_score ($r = -0.69$) are far more strongly correlated with churn_risk than any behavioural feature (review_length $r = 0.51$, thumbs_up_count $r = 0.03$). Closer inspection confirmed why: **reviews with a rating of 3, 4, or 5 are churn_risk = 0 in 100% of cases**, with essentially all label variance concentrated in ratings of 1-2, and **92.7% of reviews with negative sentiment are churn_risk = 1** versus 0% for neutral or positive sentiment. This indicates churn_risk was constructed using a rule closely tied to rating and sentiment_score, rather than being an independently observed outcome — a label leakage risk that is addressed directly in Section 4.

4. Analytics Method and Implementation

Method Selection: Supervised Binary Classification

Predicting `churn_risk` (0/1) from review features is a binary classification problem. This is an appropriate method because: (1) a clearly defined binary target variable already exists in the data; (2) a substantial labelled sample (12,800 reviews) is available for training and validation; (3) both structured features (rating, sentiment, length, reply status, app category) and unstructured text (`review_text`) are available as predictors; and (4) the business objective requires both predictive accuracy and interpretability of which signals drive risk.

Addressing Label Leakage

As identified in Section 3, `rating` and `sentiment_score` were found to near-deterministically define `churn_risk`. An initial model trained with these features included achieved an unrealistic Logistic Regression ROC-AUC of 0.998 (94% precision / 99% recall on the at-risk class) — a result too clean to reflect genuine predictive learning rather than the model simply decoding the rule used to build the label. Both features were therefore **excluded from the final model**, which instead relies only on features that are independent of how the label was generated: review text (via TF-IDF), `review_length`, `word_count`, `thumbs_up_count`, `has_reply`, and `app_category`.

Implementation

Two classifiers were trained and compared using scikit-learn, on a stratified 80/20 train-test split (10,240 train / 2,560 test rows, preserving the 14.5% churn rate in both splits):

Classifier	Rationale
Logistic Regression	Linear baseline with <code>class_weight='balanced'</code> ; interpretable coefficients for feature-level insight
Random Forest	Ensemble of 300 trees (<code>max_depth=12</code>); captures non-linear interactions and provides feature importance ranking

Structured features were standard-scaled (numeric) / one-hot encoded (categorical) and combined with a 3,000-feature TF-IDF matrix (unigrams and bigrams, `min_df=5`) built from cleaned review text, producing a combined feature space of 3,014 dimensions.

5. Comparison with Published Studies

Study	Method	Key Finding	Comparison with This Work
Ahmad, Jafar & Aljoumaa (2019), "Customer Churn Prediction in Telecom Using Machine Learning in Big Data Platform," Journal of Big Data, 6(28).	XGBoost on transactional/RFM-style telecom features (tenure, usage, contract type)	AUC of 0.93 predicting churn from account and usage history	Their model uses lagging transactional history; ours uses review text and metadata as a leading indicator available the same day a complaint is posted. Our corrected AUC (0.978 Logistic Regression) is comparable despite using an entirely different, text-based signal source.
Guzman & Maalej (2014), "How Do Users Like This Feature? A Fine-Grained Sentiment Analysis of App Reviews," IEEE RE Conference.	Lexicon and NLP-based sentiment analysis of app store reviews at the feature-mention level	Sentiment expressed toward specific app features can be reliably extracted from review text	Directly comparable data source (Play Store reviews). Our complaint_theme clustering plays a similar role to their feature-level sentiment extraction, but is applied specifically to predict churn risk rather than general feature satisfaction.
Chen, Lin, Hoi, Xiao & Zhang (2014), "AR-Miner: Mining Informative Reviews for Developers from Mobile App Marketplace," ICSE.	Topic modelling (ASUM) plus classification to separate informative from uninformative app reviews	A significant share of app reviews contain actionable developer-relevant information (bug reports, feature requests)	AR-Miner filters reviews for product-development relevance; our study instead filters/scores reviews for churn-risk relevance. Both confirm review text carries structured, minable signal beyond the star rating alone.

Discussion: Unlike Ahmad et al., who rely on structured account/usage history that is only available internally to the telecom provider, this study demonstrates that publicly visible review text can serve as a complementary, earlier-arriving churn signal for e-commerce platforms — often before any transactional drop-off is measurable. Consistent with Guzman & Maalej and Chen et al., our TF-IDF results confirm that review text carries strong, minable structure (complaint themes, sentiment-laden vocabulary) rather than being unstructured noise. A key methodological point this study surfaces, not directly discussed in the benchmarked papers, is the risk of **label leakage** when a churn label is itself constructed from rating or sentiment — underscoring the importance of feature-label independence in any review-based churn model.

6. Results, Business Insights, and Recommendations

Model Performance — Corrected (Leakage-Free) Models

After excluding rating and sentiment_score, both models were retrained on the remaining independent features (TF-IDF text, review_length, word_count, thumbs_up_count, has_reply, app_category):

Model	Accuracy	Precision (At-Risk)	Recall (At-Risk)	F1 (At-Risk)	ROC-AUC
Logistic Regression	0.94	0.71	0.93	0.81	0.978
Random Forest	0.90	0.62	0.84	0.71	0.954

Selected Model: Logistic Regression. It achieves the best balance for the business use case — a higher recall-precision trade-off on the minority (at-risk) class than Random Forest, correctly identifying 93% of genuinely at-risk reviews with 71% precision, and the highest overall ROC-AUC (0.978) among the two corrected models. For reference, the uncorrected Logistic Regression (including rating and sentiment_score) scored an inflated ROC-AUC of 0.998 — the 0.978 corrected figure is the honest, defensible result for this report, reflecting genuine

predictive signal from review text and behavioural metadata rather than a restatement of the label-construction rule.

Key Business Insights

- **Returns and refund handling is the single largest driver of churn risk**, accounting for 62.7% of all at-risk reviews — far ahead of delivery issues (17.2%) or product authenticity concerns (7.0%). This is the highest-leverage area for operational investment.
- **Churn-risk rate varies sharply by app**, from 31.1% (Amazon) down to 5.1% (Flipkart) among the 8 apps studied, indicating that churn drivers are platform-specific and should not be addressed with a one-size-fits-all policy.
- **Text content alone is strongly predictive of churn risk** even after removing rating and sentiment score: words such as "bad", "worst", "delivery", "refund", and "support" dominate at-risk reviews, confirming that complaint language carries independent signal beyond the star rating.
- **Explicit churn language is rare** — only 0.5% of reviews explicitly mention uninstalling or switching apps — meaning most churn risk must be inferred from complaint content and behaviour rather than waiting for users to announce their departure.
- **Label construction matters as much as model choice**. The initial near-perfect model (ROC-AUC 0.998) was an artifact of rating and sentiment_score being used to define the target; the corrected, leakage-free model (ROC-AUC 0.978) is a more honest and more useful signal for deployment, since it will generalise to genuinely new reviews rather than merely re-stating a known rule.

Recommendations

- Route new reviews through the corrected Logistic Regression pipeline in near-real-time to flag at-risk reviews for same-day CRM follow-up, prioritised by thumbs_up_count (visibility) and predicted probability.
- Prioritise fixing returns/refund process friction first — it is the dominant complaint theme behind churn risk and offers the largest expected retention return.
- Investigate platform-specific causes behind the wide churn-risk gap between apps (e.g. Amazon 31.1% vs. Flipkart 5.1%) rather than assuming a shared root cause.
- A/B test whether proactively replying to at-risk reviews (has_reply) measurably reduces subsequent negative sentiment or uninstall behaviour, extending this from a detection tool into a tested retention intervention.
- Re-validate the churn_risk labelling methodology periodically to ensure new labels are not re-introducing the rating/sentiment leakage identified in this study.

7. Conclusion and References

Conclusion

This case study demonstrates that app store reviews, though unstructured and secondary in nature, carry measurable, minable signals of customer churn risk. Of 12,800 reviews scraped across 8 Indian e-commerce apps, 14.5% were flagged as churn-risk, with returns and refund-handling complaints responsible for nearly two-thirds of all at-risk cases. An important methodological finding emerged during modelling: an initial classifier using rating and sentiment_score as predictors achieved a suspiciously perfect ROC-AUC of 0.998, which investigation revealed was due to label leakage — churn_risk had been constructed directly from these two fields. After excluding them, a corrected Logistic Regression model achieved a genuine ROC-AUC of 0.978 using only review text and behavioural metadata, confirming that complaint language and reviewer behaviour independently carry strong predictive signal. These findings give e-commerce businesses a faster, cheaper, and earlier churn indicator than waiting on lagging transactional purchase data alone, and directly support the objectives set out in Section 1.

References

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